

Data Structures and Algorithms

1. What is time complexity and space complexity? Explain best case, worst case and average case time complexity with examples.
2. Display a given range of numbers in reverse order (using recursion).
3. How to reverse a singly linear linked list?
4. How to reverse a singly linear linked list using recursion?
5. Display singly linear linked list in reverse order.
6. Find the middle node of a singly linear linked list.
7. Check if a singly linear list contains a loop.
8. Check if the given singly linear list is palindrome or not.
9. Create a stack (LIFO) using queue (FIFO).
10. Implement a quick sort algorithm.
11. How to solve a rat in a maze problem?
12. What is a hash table? Implement hash tables with separate chaining.
13. Input a string from the user and find the character repeated maximum number of times (irrespective of case).
14. Print the length of the highest continuous number range in the given array.
(<https://www.geeksforgeeks.org/longest-consecutive-subsequence/>)
15. Flatten a multi level doubly linkedlist (You can find this problem in Leetcode - <https://leetcode.com/problems/flatten-a-multilevel-doubly-linked-list/>)
16. Find the maximum sum for any contiguous nodes in a given linkedlist.
(<https://www.geeksforgeeks.org/maximum-sum-contiguous-nodes-in-the-given-linked-list/>)
17. Given an array arr[] of N integers, calculate the median.
(<https://leetcode.com/problems/sliding-window-median/description/>)
18. Count number of palindrome trees in given input.
(<https://www.hackerearth.com/problem/algorithm/palindrome-tree/>)